

Locally Biased, Financially Blind: Where State Investor Literacy Moderates the Momentum–Idiosyncratic-Volatility Anomaly

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Abstract

Retail investors trade locally, and they trade more like the people around them than the textbook investor: chasing recent winners, anchoring on familiar names, paying limited attention to volatility-adjusted risk. State-level financial literacy varies substantially across the United States, and a large household-finance literature documents that less literate investors make systematically worse portfolio choices. If those choices show up in prices, they should show up as a state-literacy moderation of return anomalies that depend on retail mispricing. Yet in aggregate, the empirical signature is faint: a regression

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of the momentum–idiosyncratic-volatility interaction on state literacy returns modest coefficients without a clean economic story. This paper asks a sharper question: not whether literacy moderates the anomaly, but *where in the cross-section* it does, and what microfoundation that location identifies. The answer, on the stable-headquarters subsample of CRSP-listed U.S. common stock from 2009 to 2023, is that the moderation lives in firms that occupy an interaction zone — enough retail demand to move prices, enough institutional activity to translate that demand into return differences across literate and illiterate states. In the full panel these firms fall in the middle of the institutional-ownership distribution; under sample restrictions to larger firms they migrate to the low-IO tercile, exactly as the interaction-zone reading predicts. Firms dominated by institutions show no literacy gradient at all, consistent with arbitrage neutralizing the local-state channel. The contribution is a precisely-bounded characterization of where a literacy mechanism operates, and an empirical signature — migration with size composition — that identifies the mechanism without requiring an instrumental-variables strategy.

1 Introduction

Lusardi and Mitchell [2014] and a large literature document that financially literate households make systematically better portfolio decisions: they diversify more, panic less, time the market less, and pay more attention to fees

and volatility-adjusted risk. The variation across U.S. states is substantial — roughly 28 percent of adults answer three of five basic literacy questions correctly in high-literacy states like Minnesota and New Hampshire, against around 21 percent in Mississippi and West Virginia. The local-bias literature supplies the second piece: retail investors hold portfolios tilted toward locally headquartered firms [Coval and Moskowitz, 1999, Ivković and Weisbenner, 2005], and this local demand moves prices [Hong et al., 2008, Bernile et al., 2015]. Putting the two together yields a clean hypothesis: where retail investors are less literate, they should trade more like the textbook noise trader, so state literacy should moderate the momentum–idiosyncratic-volatility (IV) interaction — a retail-friendly anomaly combined with the standard limits-to-arbitrage measure — with the effect weaker in literate states.

In aggregate, however, the data do not deliver. Regressing the three-way interaction on the full panel of CRSP-listed common stock returns a modest negative coefficient with marginal significance, and the magnitude collapses by roughly forty fold once headquarters-state assignment is corrected for the 14 percent of firm-months whose static Compustat snapshot does not match their then-current 10-K filing. The aggregate test is misleading in two ways: it pools firms with very different ownership structures, and it pools firms whose local-investor base is well-defined (stable headquarters) with firms whose base is in transition (relocators). The aggregate null is therefore not informative about whether a literacy channel exists — only about whether it operates uniformly across the cross-section. It does not.

The contribution of this paper is to ask a sharper question: where in the cross-section does the literacy mechanism become visible, under what conditions on firm ownership and size, and what does the answer identify about the microfoundation? On the stable-headquarters subsample of CRSP-listed common stock from 2009 to 2023, the literacy moderation concentrates in firms whose institutional ownership sits in the middle of the within-stock IO distribution — the mid-IO tercile, with mean ownership share around fifty percent. Firms with predominantly retail ownership (low IO) show essentially no moderation; firms with predominantly institutional ownership (high IO) also show none. The mid-tercile coefficient survives the addition of firm-size moderation as a control, so the finding is not a size effect dressed in ownership clothing even though mid-IO firms are predominantly mid-cap. Within mid-IO, the coefficient corresponds to a cross-state literacy gradient in exactly the predicted direction: higher-literacy states show weaker anomaly slopes, and the across-state correlation between per-state coefficients and state literacy is sharp and clean.

The economic interpretation the data identify is an *interaction zone*. The literacy channel requires two ingredients to leave a price footprint: retail demand large enough to move prices, and institutional activity sufficient to translate that demand into return differences detectable across states. Where neither is present (micro-cap firms held mostly by retail with no institutional counterparty), retail mispricing is genuine but accumulates uniformly without loading on cross-state literacy variation. Where institutions dominate

(large firms with high IO), arbitrage absorbs retail demand and neutralizes the local-state channel. The interaction zone — firms with mixed ownership at meaningful market size — is where both ingredients are present. In the full stable-HQ panel it falls predominantly in mid-IO; dropping the bottom 50 percent of firms by market capitalization shifts the moderation to low-IO, and the cross-state gradient migrates with it. It is the same economic population, relocated to a different tercile of the conditional IO distribution.

This migration is the central piece of mechanism evidence the paper introduces — the empirical signature an interaction-zone reading predicts and the IO-tercile result alone does not deliver. The high-IO tercile shows no cross-state gradient under any sample restriction, a clean predicted-null check the reading passes. The IV-decile decomposition within mid-IO, which would test the canonical limits-to-arbitrage microfoundation under which the effect strengthens at high IV, fails to reject decile equality and returns a near-zero coefficient at the highest IV decile. The textbook limits-to-arbitrage reading is rejected; the interaction-zone reading, identified through size and ownership conditioning rather than through IV, is identified positively by the migration.

Three bounds discipline the result. First, the continuous-margin quadratic-IO U-shape clears five percent on a quadratic specification only — not under log, kink, or twenty-bin alternatives — so the functional-form-free claim is the discrete-tercile contrast. Second, under point-in-time literacy reassignment on the full panel the mid-IO coefficient collapses by roughly twenty-six

fold; the characterization is a statement about firms with stable headquarters, not the corrected full panel. Third, an instrument using state K-12 personal-finance graduation requirements fails its first-stage relevance test in the right direction (mandate states are systematically less literate, having adopted mandates as a remedy), so the OLS estimate is the preferred inferential statement and the IV failure is documented honestly.

The closest prior work is Korniotis and Kumar [2013], who show that state-level business cycles predict returns of locally headquartered firms. This paper differs on three margins: the state variable is NFCS literacy rather than the business cycle, the object is a three-way moderation of an anomaly interaction rather than a level prediction, and the discriminating margin is a within-stock IO contrast the business-cycle channel does not predict. A business-cycle control block attenuates the literacy three-way by 20–23 percent, so the two state moderators are distinct but partly overlapping. Korniotis et al. [2020] use brokerage-demographic sophistication proxies but neither NFCS literacy nor the momentum-IV interaction; Hillert et al. [2014] is the closest precedent for a state-level moderator of momentum, using media coverage with state individualism as the cross-sectional moderator rather than literacy.

Roadmap. Section 2 describes the data, the NFCS state-literacy panel, and the stable-headquarters subsample. Section 3 states the specification and identifying assumptions. Section 4 reports the mid-IO concentration,

the size-controlled extension, the migration, and the cross-state gradient. Section 5 develops the interaction-zone characterization. Section 6 reports the disciplining battery. Section 7 discusses interpretation and limitations, and Section 8 concludes.

2 Data and Sample Construction

2.1 Sample

The sample is the cross-section of CRSP-listed common stock (share codes 10 and 11; exchange codes 1, 2, 3, covering NYSE, NYSE American, and NASDAQ) from January 2009 through December 2023. The full panel contains 623,896 firm-month observations, 7,447 unique firms, 51 headquarters states (the fifty states plus the District of Columbia), and 180 calendar months. I z-score all focal variables within each calendar month, so the residual variation the regressions exploit is cross-sectional and free of time-aggregate movements. State financial literacy is measured from the NFCS Big Five literacy questions, fielded by the FINRA Investor Education Foundation in waves 2009, 2012, 2015, 2018, and 2021, and forward-filled to monthly frequency between waves.

2.2 The stable-headquarters subsample

The local-bias channel transmits through community ties between a firm and its headquarters-state retail base, and those ties take time to form and decay, so for firms that relocate within sample the effective local-investor population is ambiguous. I drop firms with a confirmed in-sample EDGAR-based relocation, using the union of two flags — `hdr_changed` (the 10-K-header business-address `STATE` changes across a firm’s in-sample 10-Ks; 954 firms) and `sec_disagree` (the most-recent 10-K-header state disagrees with the static Compustat snapshot, with ≥ 2 parsed headers; 524 additional firms). The union identifies 1,366 relocators (16.8 percent of firm-months), and the top first-to-last state pairs are consistent with the relocation mechanism of Pirinsky and Wang [2006]. The headline patterns are estimated on the resulting stable-headquarters subsample (519,090 firm-months, 6,081 firms) under static-snapshot literacy; the point-in-time reassignment on the full panel is the correctness check (Section 6.1). Construction and PIT-coverage detail are in IA A and IA B.

2.3 Variable definitions

The dependent variable is the monthly stock return `ret` from CRSP. Momentum is `mom_12_2`, the standard Jegadeesh and Titman [1993] cumulative return from twelve months before to two months before the observation month, skipping the most recent month. Idiosyncratic volatility (IV) is the standard

deviation of daily Fama-French three-factor residuals over a rolling twelve-month window. State financial literacy is the NFCS Big Five share-correct score, z-scored within month and denoted `literacy_z`.¹ Each cross-product (the two-way terms $\text{mom} \times \text{IV}$, $\text{mom} \times \text{literacy_z}$, $\text{IV} \times \text{literacy_z}$, and the three-way $\text{mom} \times \text{IV} \times \text{literacy_z}$) is z-scored within month. The Fama-French 12-industry block and the Korniotis and Kumar [2013] business-cycle block enter robustness specifications.

2.4 Institutional ownership from the Thomson Reuters s34 13F panel

I build a firm-quarter institutional-ownership (IO) panel from the WRDS Thomson Reuters s34 13F holdings file for 2009Q1 through 2023Q4 (222,180 firm-quarter observations, 7,907 distinct permnos). The numerator follows the literature-standard convention — amendment-deduplicated filings (latest filing per manager-security-report-date triple) and the sole-plus-shared investment-discretion measure rather than the raw `shares` field — which brings the residual above-one rate to 0.36 percent (vs 6.49 percent under raw `shares`), capped at one. Full construction detail is in IA P.

The persistent IO measure is each firm’s time-mean institutional-ownership share across its available Thomson quarters; the time-varying measure assigns each firm-month the share from the most recent 13F quarter with report date

¹Correct-answer keys M6=1, M7=3, M8=2, M9=2, M10=1, verified from the FINRA Foundation NFCS State-by-State Codebook (2021); full construction detail in Appendix A.

Table 1: Descriptive statistics and state literacy rankings
Panel A: Distribution of focal variables (full panel)

Variable	Mean	SD	p05	p50	p95
Monthly return	0.013	0.192	−0.220	0.005	0.253
Momentum (<code>mom_12_2</code>)	0.107	0.828	−0.678	0.034	1.021
Idiosyncratic volatility	0.027	0.022	0.007	0.020	0.070
Market equity (\$ billions)	7.04	42.1	0.017	0.596	26.19
NFCS Big Five share correct	0.244	0.028	0.200	0.247	0.286

Panel B: Panels and merge conventions

Panel	Firm-months	Firms
Full panel (static-snapshot HQ, baseline literacy merge)	623,896	7,447
Full panel, PIT literacy reassignment	623,814	7,447
Stable-HQ subsample (no in-sample EDGAR relocation)	519,090	6,081
with persistent IO measure	518,962	6,018
with time-varying IO measure	493,889	5,877

Panel C: State literacy rankings and NFCS wave summary

Highest-literacy states	MN, HI, NH, DC, CT
Lowest-literacy states	MS, OH, AR, TX, WV
NFCS wave means (2009, 2012, 2015, 2018, 2021)	0.258, 0.254, 0.255, 0.229, 0.222
Median within-state rank correlation across waves	0.71 (Spearman)

Notes. Idiosyncratic volatility is the standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window. Market equity is price times shares outstanding from CRSP, in billions of dollars. The NFCS Big Five share-correct score is the state-wave share of respondents answering the FINRA Big Five financial-literacy questions correctly. The within-state rank correlation across the five NFCS waves is the median Spearman correlation of state-wave ranks across consecutive waves (a measurement-stability statistic for the state literacy ordering).

at least four months earlier (a strictly conservative no-look-ahead merge). On the stable-headquarters subsample, persistent IO covers 518,962 firm-months and time-varying IO covers 493,889 firm-months.

2.5 Descriptive statistics

Table 1 reports the cross-sectional distribution of the focal variables.

3 Identification

The introduction posed two questions: where in the cross-section the literacy mechanism becomes visible, and what microfoundation that location identifies. This section describes the design that answers them.

3.1 What the research question requires

Testing whether state literacy moderates the momentum–IV anomaly — more strongly in literate states — means measuring how the within-stock relationship between returns and $\text{mom} \times \text{IV}$ varies with the headquarters-state literacy of each stock. Three features of the data dictate the design. First, literacy is measured at the state-wave level (the FINRA NFCS fields the Big-Five questions every three years, five wave values over 2009–2023), while anomaly returns are at the firm-month level; bridging the scales requires the identifying variation to be cross-firm *within* state-month cells, not cross-state (which would conflate literacy with other state-level return determinants) and not cross-time (which would conflate it with aggregate dynamics). Second, the channel transmits through local-investor community ties that take time to form and decay, so for firms that relocate headquarters the effective local base is in transition; a clean design restricts to stable-headquarters firms and reports the static-vs-point-in-time correction as a falsification check. Third, the prediction concerns how the moderation varies with ownership structure, so the design stratifies by within-stock institutional-ownership tercile rather

than collapsing across IO levels.

3.2 The baseline specification

The headline specification implements these requirements as a two-way fixed-effects panel regression:

$$\begin{aligned}
r_{i,s,t} = & \alpha_s + \tau_t + \gamma (\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{literacy_z}_{s(i,t),t}) \\
& + \delta_1 (\text{mom}_{i,t} \cdot \text{IV}_{i,t}) + \delta_2 (\text{mom}_{i,t} \cdot \text{literacy_z}_{s(i,t),t}) + \delta_3 (\text{IV}_{i,t} \cdot \text{literacy_z}_{s(i,t),t}) \\
& + \beta' \mathbf{X}_{i,s,t} + \varepsilon_{i,s,t}.
\end{aligned} \tag{1}$$

The dependent variable $r_{i,s,t}$ is the month- t return of firm i headquartered in state $s(i, t)$. State fixed effects α_s absorb time-invariant state characteristics; month fixed effects τ_t absorb aggregate shocks. The three two-way interactions enter as controls; the three-way coefficient γ is the object of interest. The state-assignment function $s(i, t)$ is the static Compustat snapshot under the headline merge and the point-in-time EDGAR 10-K-header reassignment under the falsification check; on the stable-headquarters subsample the two coincide. The discrete-tercile rendering of Equation (1) re-estimates γ within each within-stock IO tercile.

3.3 Adding size as a moderator: why and how

Mid-IO firms are concentrated in mid-cap market-equity quintiles (median market capitalization roughly five to ten times higher than in low-IO), so a reader could ask whether the mid-IO three-way is a size moderation captured by the IO discretization. The size-controlled specification rules this out by entering size as a parallel moderator on the same anomaly:

$$\begin{aligned}
 r_{i,s,t} = & \alpha_s + \tau_t + \gamma \left(\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{literacy_z}_{s(i,t),t} \right) \\
 & + \zeta \left(\text{mom}_{i,t} \cdot \text{IV}_{i,t} \cdot \text{size_z}_{i,t} \right) \\
 & + \text{lower-order literacy interactions} + \text{lower-order size interactions} + \varepsilon_{i,s,t},
 \end{aligned}
 \tag{2}$$

where $\text{size_z}_{i,t}$ is the within-month z -score of $\log(\text{ME}_{i,t})$, entered with the same standardization and interaction structure as literacy. The coefficient γ in Equation (2) is identified holding size moderation of the same anomaly fixed: if the literacy moderation is mechanically conflated with size through the mid-IO sample's mid-cap concentration, γ should attenuate to insignificance; if literacy is a distinct moderator, γ should survive. Equation (2) is the preferred specification throughout.

3.4 Identifying assumptions

For γ to be interpretable as a literacy moderation, three assumptions must hold.

Conditional unconfoundedness for the literacy moderator. Conditional on state and month fixed effects, the three two-way interactions, and the size moderator in Equation (2), residual variation in state-aggregate literacy is uncorrelated with omitted return determinants that interact with $\text{mom} \times \text{IV}$ in the same functional form. The paper addresses this four ways: a Fama-French 12-industry control block; a state college-attainment horse-race within mid-IO (literacy three-way -0.0350 , $t = -2.97$; college $+0.00076$, $t = +0.65$); the within-stock IO contrast, which discriminates against any generic state-level moderator hitting all headquarters-state stocks alike; and the size-controlled specification, which discriminates against the most natural firm-level moderator. What remains — a class of ownership-interactive state-level confounds the design does not separate — is engaged in Section 7.

Correct headquarters-state assignment. The static Compustat snapshot mis-assigns relocater firm-months for the 14 percent of observations whose static state differs from the then-current 10-K headquarters state. Headline patterns are estimated on the stable-HQ subsample, for which the static and point-in-time merges coincide; the PIT reassignment on the full panel is the strictest available correction, and the resulting magnitude col-

lapse (Section 6) is the binding population-scope limitation.

Cluster-robust inference with 51 state clusters. The natural clustering dimension is the state, and with 51 clusters asymptotic cluster-robust standard errors over-reject; the binding procedure is the one-way state-level wild-cluster bootstrap of Cameron et al. [2008] with Webb six-point weights [Webb, 2014] and the null imposed.

3.5 Causal language

Under conditional unconfoundedness, $\hat{\gamma}$ in Equation (2) recovers the average within-state-month differential sensitivity of returns to $\text{mom} \times \text{IV}$ as a function of state-aggregate literacy; it is the conditional association the design supports, not a clean average treatment effect.²

4 Results

The introduction asked where in the cross-section the literacy mechanism becomes visible. This section answers in four steps: the moderation concentrates in mid-IO firms and survives size as a competing moderator; the

²I attempted an instrumental-variables strategy using state K-12 personal-finance graduation requirements as Z for state literacy (building on Urban-Schmeiser 2015/2020). The instrument fails its first-stage relevance test for documented reasons — mandate adoption was a state-level response to low adult literacy, producing a wrong-signed cross-state correlation — and a cohort-matched implementation with state, wave, and age-band fixed effects yields first-stage $F = 0.05$. 2SLS estimates are directionally consistent with OLS but with standard errors inflated 6–7 $\times$; the OLS estimate is the preferred inferential statement. Full detail is in IA V and Section 6.

finding is a property not of mid-IO firms specifically but of an underlying economic population whose IO location shifts with the sample’s size composition (the central mechanism evidence); that same population carries a cross-state literacy gradient in the predicted direction while firms outside it show none; and a parametric U-shape and a sample-selection decomposition close the section.

4.1 Where the moderation lives: a U-shape in institutional ownership

To see whether literacy moderates the momentum–IV interaction differently across firms, I discretize the within-stock IO distribution into terciles and estimate Equation (1) within each on the stable-HQ subsample. The naive expectation is that the channel is strongest where retail dominates (low IO) and fades where institutions dominate (high IO), with the middle tercile ambiguous. Table 2 reports the result under persistent and time-varying IO measures.

The naive prediction is rejected. The low-IO tercile — the most retail-driven and supposedly most literacy-sensitive — returns a null coefficient, and the high-IO tercile is also null, consistent with institutional dominance neutralizing the local-state channel. The moderation concentrates in the middle: mid-IO is the only tercile with state-clustered $|t| > 2$ on both IO measures, with magnitude roughly three times the low-IO and six times the

Table 2: Mid-IO concentration: three-way coefficient by IO tercile, stable-HQ subsample

<i>Panel A: Persistent IO measure</i>					
Tercile	Mean IO share	$\hat{\gamma}$	State-clustered t	wcb p	N
IO1 (low; high retail)	0.170	−0.0113	−1.04	0.296	144,989
IO2 (mid)	0.495	−0.0321	−2.63	0.064	191,589
IO3 (high; institutional)	0.704	−0.0050	−0.31	0.760	182,384
<i>Panel B: Time-varying IO measure</i>					
Tercile	Mean IO share	$\hat{\gamma}$	State-clustered t	wcb p	N
IO1 (low; high retail)	0.174	−0.0087	−0.92	0.337	141,129
IO2 (mid)	0.504	−0.0265	−2.28	0.058	184,881
IO3 (high; institutional)	0.711	−0.0044	−0.25	0.789	167,879

Notes. Each row reports the three-way coefficient $\hat{\gamma}$ on $\text{mom} \times \text{IV} \times \text{literacy_z}$ estimated within the IO tercile, on the stable-HQ subsample under static-snapshot literacy. The persistent measure is each firm’s time-mean institutional-ownership share; the time-varying measure assigns each firm-month the share from the most recent 13F quarter with report date at least four months earlier. Wild-cluster bootstrap uses Webb six-point weights [Cameron et al., 2008] with $B = 4,999$ and null imposed. Sample: January 2009 through December 2023.

high-IO estimates. This functional-form-free discrete contrast is the load-bearing characterization; Section 4.5 reports a continuous quadratic specification that corroborates the U-shape but does not replace it.

4.2 Is the U-shape just a size effect?

Mid-IO firms are also concentrated in the middle of the market-capitalization distribution (median ME \$561 million, versus \$66 million for low-IO and \$1.40 billion for high-IO), so a reader could wonder whether the finding is a size effect captured by the IO discretization. The size-controlled specification (Equation 2) tests this by entering size as a parallel moderator on the same

Table 3: Size-controlled mid-IO three-way: literacy survives

IO tercile	Baseline $\hat{\gamma}$ (state- t)	Size-controlled $\hat{\gamma}$ (state- t)	N
IO1 (low)	-0.0113 (-1.03)	-0.0077 (-0.65)	144,989
IO2 (mid)	-0.0321 (-2.61)	-0.0256 (-2.24)	191,589
IO3 (high)	-0.0050 (-0.31)	-0.0001 (-0.00)	182,384
Horse race in mid-IO: γ_{size} on $\text{mom} \cdot \text{IV} \cdot \text{size_z}$	—	+0.0017 (+0.64)	191,589

Notes. Baseline column re-estimates Equation (1) within IO tercile. Size-controlled column estimates Equation (2) within IO tercile, adding **size_z**, $\text{mom} \cdot \text{size_z}$, $\text{IV} \cdot \text{size_z}$, and $\text{mom} \cdot \text{IV} \cdot \text{size_z}$. State and month fixed effects throughout. State-clustered t in parentheses. The horse-race row reports the coefficient on size's three-way under the size-controlled specification within mid-IO.

anomaly, with the same standardization and interaction structure as literacy: if the mid-IO three-way is mechanically conflated with size, the literacy coefficient should attenuate to insignificance; if literacy is distinct, it should survive. Table 3 reports the comparison.

Literacy survives. The mid-IO three-way attenuates about twenty percent (to -0.0256) but stays significant at $t = -2.24$, while size's own three-way within mid-IO is statistically zero: conditional on being inside the mid-IO tercile, the literacy moderation is identified independently of size. The specification also sharpens the U-shape, turning the already-small high-IO coefficient into a clean null (-0.0001 , $t = -0.00$).

4.3 The migration test: what moves when the sample changes

The size-controlled result tells us the moderation lives in mid-IO but not *why*. A natural reading is that mid-IO is simply where the full panel's size

Table 4: Interaction-zone migration: IO tercile three-way coefficient under size filters, size-controlled spec

Panel A: Within-panel three-way coefficient $\hat{\gamma}_{lit}$ by IO tercile

Size filter	IO1 (low)	IO2 (mid)	IO3 (high)	N
Full stable-HQ panel (paper headline)	-0.0077 (-0.65)	-0.0256 (-2.24)	-0.0001 (-0.00)	518,962
Drop bottom 20% (micro-cap)	-0.0137 (-1.14)	-0.0402 (-2.57)	+0.0109 (+0.75)	454,107
Drop bottom 33% (small-cap)	-0.0155 (-1.02)	-0.0273 (-1.29)	+0.0122 (+0.66)	395,873
Drop bottom 50% (below median)	-0.0441 (-2.04)	-0.0098 (-0.58)	+0.0115 (+0.58)	322,137

Panel B: Cross-state weighted slope of per-state $\hat{\gamma}_{lit}$ on state literacy

Sample cell	Weighted slope	state- t	Pearson r	n_{states}
Full panel, mid-IO (headline cell)	-0.0623	-5.01	-0.275	42
Drop bottom 50%, mid-IO	-0.0325	-3.82	+0.008	31
Drop bottom 50%, IO1 (migrated headline cell)	-0.0379	-4.30	-0.291	34
Drop bottom 50%, IO3 (predicted null)	+0.0061	+0.26	+0.054	31

Notes. Panel A: Equation (2) within each IO tercile, under each size filter applied at the firm level using firm panel-mean market capitalization. State and month fixed effects throughout. State-clustered t in parentheses. Bold cells indicate the tercile carrying the literacy moderation under each filter. Panel B: per-state $\hat{\gamma}_{lit}$ from Equation (2) with month fixed effects only (state fixed effects collapse for single-state regressions); the cross-state regression of $\hat{\gamma}_{lit}$ on state NFCS literacy uses precision weighting ($1/SE^2$). States with fewer than 5 firms in the sample cell are excluded. The cross-state state- t reported is the precision-weighted-slope t -statistic.

composition puts the interaction zone, so the moderation should follow that zone as the sample changes rather than being a fixed property of the tercile. Restricting to larger firms removes the micro-caps that populate low-IO and shifts the conditional IO distribution upward; under the interaction-zone reading, firms that occupied mid-IO at the full-panel margin should reappear at low-IO in the restricted sample, carrying the moderation with them.

Table 4 tests this: Panel A reports the within-panel three-way coefficient under progressive size filters, and Panel B the cross-state literacy gradient in the relevant cell under each filter.

The prediction holds cell by cell. In the full panel the moderation lives in mid-IO; dropping the bottom twenty percent of firms by market cap *strengthens* the mid-IO coefficient (the micro-caps removed fall predominantly in low-IO); and dropping the bottom fifty percent relocates the significant coefficient entirely to low-IO ($\hat{\gamma}_{IO1} = -0.0441$, $t = -2.04$), with mid-IO now null. The within-panel finding is not a property of mid-IO firms but of an underlying population whose IO label depends on what else is in the sample. The cross-state gradient migrates with it: it lives in mid-IO in the full panel ($r = -0.28$, $t = -5.01$) and in low-IO in the above-median sample ($r = -0.29$, $t = -4.30$), while high-IO shows none under either composition — the predicted-null check. The migration test is the empirical signature an interaction-zone microfoundation predicts and the IO-tercile result alone does not deliver.

4.4 Across states, does the moderation track literacy?

A direct cross-sectional test asks whether, across firms in the interaction-zone cell, the moderation tracks state literacy in the predicted direction. I estimate the three-way coefficient state by state under Equation (2) restricted to that cell, then regress the per-state coefficients on state NFCS literacy with precision weighting:

$$\hat{\gamma}_{\text{lit},s} = \alpha + \beta \cdot \text{literacy}_s + u_s, \quad \hat{\beta} = -0.0623, \text{ SE} = 0.0124, t = -5.01, \text{ Pearson } r = -0.275. \quad (3)$$

The gradient runs in the predicted direction: higher-literacy states show more negative per-state coefficients, with a state-permutation two-sided $p = 0.0047$. Across the three jointly relevant tests — the headline coefficient, the predicted-null high-IO coefficient, and the cross-state slope — Holm step-down family-wise p -values are 0.05, 0.014, and a decisive 0.996 respectively: the headline and the gradient survive multiplicity correction, and the high-IO cell is unambiguously empty.

The same negative slope appears in two settings — the full-panel mid-IO cell and the above-median IO1 cell — recovering essentially the same gradient across two IO terciles in two sample restrictions (IA Y), as the migration story predicts. Figure 1 consolidates the migration evidence: the within-panel coefficients across the four size filters, and the cross-state gradient scatters for the original headline cell, the cell that loses the gradient when smaller firms are removed, and the migrated headline cell. The interaction zone moves with the sample, and the gradient moves with it.

4.5 A continuous corroboration of the U-shape

The discrete-tercile contrast is the headline result. A natural follow-up asks whether a continuous functional form on IO recovers the same non-monotone

Figure 2: Interaction-zone migration across sample size restrictions
The mid-IO finding is sample-conditional. The underlying object is the size $\times$ ownership interaction zone.

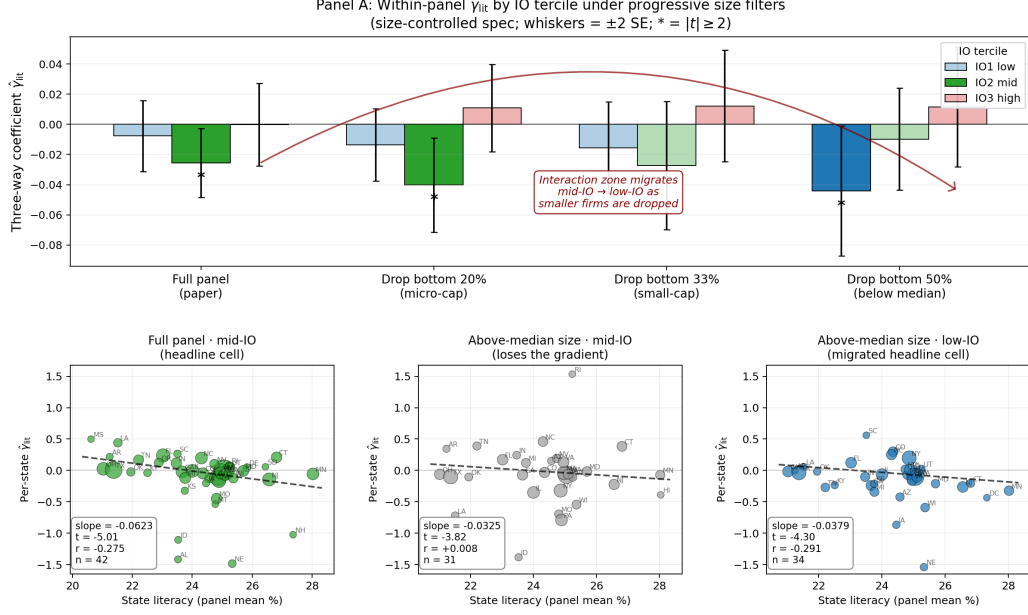


Figure 1: Interaction-zone migration across sample size restrictions. Panel A: within-panel $\hat{\gamma}_{lit}$ by IO tercile under progressive size filters, size-controlled spec, whiskers = ± 2 SE, * = $|t| \geq 2$. Panel B: cross-state scatter of per-state $\hat{\gamma}_{lit}$ on state literacy for three sample cells. Point size $\propto \sqrt{n_{\text{firms}}}$.

shape with a separable interior trough. Augmenting Equation (1) with linear and quadratic IO interactions produces the test:

$$r_{i,s,t} = \alpha_s + \tau_t + \gamma_4 (\text{mom} \cdot \text{IV} \cdot \text{literacy} \cdot \text{z} \cdot \text{IO}) + \gamma_5 (\text{mom} \cdot \text{IV} \cdot \text{literacy} \cdot \text{z} \cdot \text{IO}^2) + \beta' \mathbf{X}_{i,s,t} + \varepsilon_{i,s,t}, \quad (4)$$

where $\mathbf{X}$ collects the 21 lower-order interactions and the literacy slope $s(\text{IO}) = \gamma_3 + \gamma_4 \text{IO} + \gamma_5 \text{IO}^2$ has an interior minimum at $\text{IO}^* = -\gamma_4/(2\gamma_5)$ when $\gamma_5 > 0$. On the stable-HQ persistent measure $\hat{\gamma}_5 = +0.0151$ ($t = +2.28$, wcb

$p = 0.017$), with implied trough $IO^* = 0.344$ — inside the mid-IO tercile, exactly where the discrete contrast predicts. But the clearance rests on a single functional form: Section 6 documents $p = 0.75$ under log-IO, $p = 0.24$ under kink-IO, and a flat twenty-bin diagnostic. The U-shape exists but only the discrete-tercile contrast is robust to functional form; the quadratic is corroboration, not a separate inferential statement.

4.6 How much of the contrast comes from sample selection?

The headline drops relocators, so part of the mid-vs-low contrast reflects the firms excluded. The mid/low coefficient ratio is 2.83 on the stable-HQ panel against 1.09 on the full panel including relocators. A firm-by-firm decomposition (IA G) attributes about one-third of the gap to suppression of the low-IO comparator under the relocator drop (relocators are nearly twice as concentrated in low-IO, with a near-zero relocator-only low-IO coefficient) and about two-thirds to strengthening of mid-IO itself. Relocators are smaller, higher-IV, lower-IO, lower-return firms, consistent with a sample-selection contribution to the headline magnitude. Section 6 engages the binding population-scope question — the magnitude collapse under point-in-time literacy reassignment on the full panel — as the final disciplining check.

5 Mechanism: The Interaction-Zone Characterization

Section 4 documented four facts: the moderation concentrates in mid-IO ($\hat{\gamma}_{\text{mid}} = -0.0256$, $t = -2.24$); it relocates to low-IO when the bottom 50 percent of firms by market cap are dropped ($\hat{\gamma}_{\text{IO1}} = -0.0441$, $t = -2.04$); the per-state coefficient tracks state literacy in the cell that carries the moderation ($r = -0.28$ full-panel mid-IO, $r = -0.29$ above-median IO1) but not in high-IO ($r = -0.07$); and the within-mid-IO IV-decile decomposition fails to reject decile equality ($\chi^2 p = 0.94$), with decile 10 ≈ 0 . This section develops the interaction-zone characterization those facts identify and discusses what the design does and does not separate.

5.1 The interaction-zone reading

The moderation lives where two conditions hold at once: retail demand large enough to move prices, and institutional activity sufficient to translate that demand into cross-state return differences. Where neither holds (micro-caps held mostly by retail), mispricing accumulates but does not load on cross-state literacy variation; where institutions dominate (large high-IO firms), arbitrage absorbs retail demand and the marginal investor is not a local trader. The interaction zone — mid-cap firms with mixed ownership — is where both hold. Its IO label is sample-conditional: under the full-panel size composition it falls in mid-IO (mid-IO firms being predominantly mid-cap), and under the

above-median composition it migrates to low-IO (“low IO” among large firms meaning retail-tilted at substantial size). Table 4 traces this cell by cell; the high-IO null under both compositions is the predicted-null check the reading passes, and the IV-decile null within mid-IO (Section 5.2) is consistent, since a size-and-ownership channel predicts no IV-monotonicity.

5.2 What the IV-decile null rules out

The canonical microfoundation in the limits-to-arbitrage literature [Shleifer and Vishny, 1997, Pontiff, 2006] predicts that arbitrage costs scale with idiosyncratic risk, so literacy-modulated retail demand should displace more mispricing where arbitrage is more expensive. Within mid-IO, the within-month NYSE IV-decile decomposition tests this prediction directly (full decile table in IA X), and rejects it. Decile 10 — the highest-IV decile, where the canonical microfoundation predicts the strongest effect — returns $\hat{\gamma} = -0.035$ ($t = -1.27$), not significant and far from monotone, and a joint equality test across deciles fails to reject ($\chi^2(9)$, $p = 0.94$). The largest negative coefficient sits in decile 7, and the across-decile pattern is non-monotone; under the joint specification the Spearman $\rho(\text{decile}, \hat{\gamma}_d) = -0.67$ ($p = 0.033$) is a directional decline with IV, but with no concentration at the highest IV decile. The IV-decile margin is informative for what it rules out (canonical limits-to-arbitrage) but does not positively identify the interaction-zone reading; that identification comes from the size-restriction migration of Section 4.3.

5.3 High-IO null and college horse-race

The high-IO cross-state null ($\hat{\beta}_{\text{IO3}} = +0.018$, $t = +1.22$, $r = -0.07$; Holm $p_{\text{FWER}} = 0.996$) is the second confirmatory pattern the reading anticipates: high-IO firms are held by nationally diversified institutions indifferent to headquarters state, so state-level literacy variation says nothing about the marginal investor, and whatever retail demand exists is arbitrated away. Two diagnostics rule out the obvious alternatives. Literacy is not a proxy for education — a state college-attainment horse-race within mid-IO leaves literacy sharp ($\hat{\gamma}_{\text{lit}} = -0.0350$, $t = -2.97$) and college null ($+0.00076$, $t = +0.65$) despite a $\rho = 0.62$ state-level correlation — and the mid-IO concentration is not a variance artifact (mid-IO has lower identifying variation than low-IO yet a larger slope; Section 6).

5.4 What the design does not separate

The interaction-zone characterization is identified through the size-restriction migration and the cross-state gradient, but it does not separate three within-family microfoundations that share the same size-ownership conditioning: literacy-modulated retail demand *intensity* (the reading the paper most naturally adopts, with the cross-state gradient direction consistent with it), literacy-modulated retail *attention* (less momentum-following demand pressure in high-literacy states), and literacy-modulated retail *timing* (a clientele-timing channel generating the same gradient). The three make different

predictions for retail trade-flow data (volume, turnover, cross-spectral signatures) not available in the current panel; the TAQ-based Boehmer-Jones-Zhang retail trade indicator would separate them, a natural next-paper extension.

5.5 Population-scope limitation from the PIT correction

The characterization holds on the stable-HQ subsample, on which the local-investor unit of analysis is held fixed. Under the point-in-time 10-K-header reassignment, the aggregate three-way collapses roughly thirty-nine fold ($-0.0117 \rightarrow -0.0003$) and the mid-IO coefficient twenty-six fold ($-0.0321 \rightarrow -0.00121$): the directional within-tercile pattern survives but the magnitudes do not. The interaction-zone characterization is therefore a statement about stable-headquarters firms, not the PIT-corrected full panel. Section 6 reports the full PIT diagnostic.

6 Robustness

This section reports the disciplining battery. The point-in-time (PIT) headquarters correction that bounds the population scope of the result is the binding check and is reported in full; the remaining legs — functional-form sensitivity, the pre/post-2015 split, the failed instrumental-variables strategy, the within-retail aggregation-artifact null, the variance-channel diagnos-

Table 5: The nested baseline: aggregate and mid-IO three-way under PIT correction

<i>Panel A: Aggregate three-way coefficient</i>				
Specification	$\hat{\gamma}$	State- t	wcb p	N
Full panel, static-snapshot literacy	−0.0117	−2.17	0.078	623,896
Full panel, PIT literacy reassignment	−0.0003	−0.86	0.517	623,814
Stable-HQ subsample, static literacy	−0.0145	−1.94	0.122	519,090
<i>Panel B: Mid-IO coefficient under PIT, with and without size controls</i>				
Specification	$\hat{\gamma}_{\text{mid-IO}}$	State- t	Collapse vs. baseline	N
Stable-HQ static, persistent (paper headline)	−0.0321	−2.63	—	191,589
Stable-HQ static, size-controlled	−0.0256	−2.24	—	191,589
PIT full panel, persistent	−0.00121	−2.07	$\approx 27\times$ smaller	191,589
PIT full panel, size-controlled	−0.00100	−1.75	$\approx 26\times$ smaller	191,589

Notes. Static-snapshot merge pairs each firm-month with the literacy of its Compustat **state**; PIT reassignment re-merges against the firm’s then-current 10-K-header state and re- z -scores within month. State-level wild-cluster bootstrap (Webb six-point, $B = 4,999$ or $9,999$, null imposed). The PIT magnitude collapse is essentially the same with and without size controls ($27\times$ vs. $26\times$).

tic, the inference procedure, and the wave-anchor and vintage checks — are summarized here with full detail in the Internet Appendix.

6.1 The PIT correction and the nested-baseline collapse

The seed prior to this paper’s revisions predicted a negative aggregate three-way on the full panel and a negative within-stock retail-versus-institutional difference. Table 5 reports both objects under the static-snapshot literacy merge and under the point-in-time EDGAR 10-K-header literacy reassignment.

Two readings of the collapse compete on the margins this design supports.

A *mechanism-strength reading* says the channel operates most strongly on stable-HQ firms, where the local-investor unit of analysis is well-defined, and that the corrected full panel — including the 14% of firm-months with relocated headquarters — dilutes the average. A *sample-selection reading* says the stable-HQ subsample is selected toward firms with stronger anomaly moderation for reasons orthogonal to literacy. The relocators characteristics in IA G (relocators are smaller, higher-IV, lower-IO, lower-return firms) lean toward the sample-selection reading; the mechanism-strength reading is not defensible on current evidence. The interaction-zone characterization is therefore a statement about the stable-HQ population, and generalization to the PIT-corrected full panel is the principal population-scope limitation.

6.2 Time variation in the cross-state gradient

The within-panel mid-IO three-way coefficient is essentially identical pre- and post-2015 ($\hat{\gamma}^{\text{pre}} = -0.024$, $\hat{\gamma}^{\text{post}} = -0.026$; difference -0.002 , $t = +0.06$): the within-panel direction is stable. What varies across time is the *cross-state* literacy gradient that interprets the result through the literacy mechanism. The pooled gradient is right-signed and sharp (-0.0623 , state- $t = -5.01$); the post-2015 sub-period is sharper (-0.0706 , $t = -3.50$); the pre-2015 sub-period is wrong-signed ($+0.0403$, $t = +1.89$) and concentrated in NFCS Wave 2009, the first administration of the Big-Five module (Wave-2009 slope $+0.281$; Waves 2012–2021 directionally negative). The sub-period table and a time-variation figure (rolling-window coefficient negative in every window;

pre/post and size-filter panels) are in IA W and IA Y. The most parsimonious explanation is NFCS measurement maturation in the 2015+ waves rather than a 2015-era economic regime change: institutional ownership of the panel *rose* post-2015 (mean $\Delta \text{io_share} = +0.083$), ruling out a retail-trading regime shift, and the state-level change in IO is uncorrelated with state literacy ($r = +0.04$).

6.3 The remaining disciplining battery

The remaining legs are reported in full in the Internet Appendix and summarized here.

Functional form. The continuous-margin quadratic-IO five-percent clearance ($\hat{\gamma}_5 = +0.0151$, $p = 0.017$) does not survive log-IO ($p = 0.75$), kink-IO at the empirical minimum ($p = 0.24$), or a flat twenty-bin diagnostic; the load-bearing characterization is the functional-form-free discrete-tercile contrast of Table 2. The grid is IA U.

Instrumental variables. An attempt to instrument state literacy with K-12 personal-finance graduation requirements fails its first-stage relevance test in the right direction: the strongest candidate has $r = -0.355$ (wrong sign) and $F = 7.06$, because low-literacy regions adopted mandates as a remedy, and a cohort-matched implementation with state, wave, and age-band fixed effects returns first-stage $F = 0.05$. 2SLS is directionally consistent with OLS but with standard errors inflated 6–7 $\times$; OLS from Equation (2) is the preferred inferential statement. Full first-stage and 2SLS detail is IA V.

Within-retail null, variance channel, and education. Within the stable-HQ low-IO tercile, the low-minus-high-literacy difference in the three-way is wrong-signed ($+0.0033$, $t = +2.80$) but decomposes to ≈ 0 across within-month IV deciles — an observation-weighting aggregation artifact that provides no positive evidence against the naive sophistication reading (IA C). The mid-IO concentration is not a variance artifact (mid-IO has *lower* identifying variation than low-IO yet a larger slope), and a state college-attainment horse-race within mid-IO leaves literacy sharp (-0.0350 , $t = -2.97$) and college null ($+0.00076$, $t = +0.65$): literacy is not a proxy for education.

Inference, wave-anchor, and vintage. One-way state clustering on 51 clusters via the wild-cluster bootstrap of Cameron et al. [2008] with Webb six-point weights [Webb, 2014] and the null imposed is the binding procedure throughout.³ Restricting to the five NFCS wave-anchor years sharpens the mid-IO coefficient (-0.070 , $t = -2.09$), so the headline is not a forward-fill artifact; fixing each firm’s IO tercile to its 2009 assignment attenuates the mid/low ratio from 2.83 to 2.09 but preserves the concentration.

³The cross-state gradient is interpreted alongside a state-permutation p -value ($B = 9,999$), with Holm step-down family-wise correction across the three jointly relevant tests; Romano-Wolf step-down is a weakly more conservative upper bound. The two-way state-by-month CGM statistic is reported as a diagnostic in IA I.

7 Discussion

7.1 What the characterization changes

The characterization positions state NFCS Big-Five literacy as a cross-sectional asset-pricing moderator operating through an interaction-zone microfoundation, identified through the size $\times$ ownership margin rather than the IV margin the canonical limits-to-arbitrage reading predicts. Where household finance has studied literacy as a predictor of portfolio quality, participation, and resilience [Lusardi and Mitchell, 2014, Calvet et al., 2007, Grinblatt et al., 2012], this paper makes state-aggregate literacy a moderator on a specific anomaly interaction with precisely characterized population scope, and adds it to the local-return-predictability program of Korniotis and Kumar [2013] alongside the business-cycle channel — distinct (identified through IO and size conditioning the business cycle does not predict) but partly overlapping (a business-cycle control block attenuates the literacy three-way 20–23 percent).

Taking the size-controlled mid-IO coefficient $\hat{\gamma}_{\text{mid-IO}} = -0.0256$, the economic magnitude is a literacy-modulated annualized return differential of about -5.5% at a 0.187-standard-deviation swing in the standardized composite; the non-size-controlled coefficient implies -7% , and a classical errors-in-variables correction (reliability ratio $\lambda = 0.420$) extends the bracket to roughly -13% . This -5.5% to -13% bracket overlaps the $\approx 5\%$ state-business-cycle strategy of Korniotis and Kumar [2013].

7.2 Limitations

Population scope and sample selection. On the PIT-corrected full panel the mid-IO coefficient collapses by ≈ 26 fold, and the relocater characteristics (IA G) lean toward a sample-selection rather than mechanism-strength reading of the collapse: relocaters are smaller, higher-IV, lower-IO, lower-return firms, and about one-third of the stable-HQ mid/low gap comes from low-IO attenuation under the relocater drop. The characterization is a statement about firms with stable headquarters across the panel window and does not generalize without modification to the corrected full panel. Pre/post-relocation event studies on the 1,366 dropped firms would discriminate between the two readings.

Microfoundation and time variation. The characterization is positively identified through the size-restriction migration and the cross-state gradient but does not separate the within-family readings (demand intensity, attention, timing) that share the same size $\times$ ownership conditioning; the TAQ-based Boehmer-Jones-Zhang retail trade indicator would separate them. Separately, the pooled cross-state gradient (Holm FWER $p = 0.014$) is driven primarily by post-2015 data; the pre-2015 slope is wrong-signed and concentrated in NFCS Wave 2009. The most likely explanation is measurement noise in the first Big-Five administration rather than a regime change — institutional ownership rose post-2015 and is uncorrelated with the literacy gradient — and the within-panel mid-IO coefficient is essentially identical

pre- and post-2015, so this is a power issue, not a substantive shift.

Identification: IV, confounds, and functional form. The K-12 personal-finance mandate instrument is endogenous to state literacy through reverse causality (low-literacy states adopted mandates as a remedy) and fails its first-stage relevance test; the OLS estimate is the preferred inferential statement, and a clean IV would require pre-1990 mandate variation [Bernheim et al., 2001] or verifiable-compliance cohort micro-data (IA V). The within-stock contrast does not discriminate against an ownership-interactive state-level confound; the canonical instance is state-resident public-pension home-state ownership (≈ 21 managers in roughly seven states, collinear with the state fixed effect elsewhere), whose closure requires a CIK-by-CIK state-pension-resident flag not built here. Finally, the continuous-margin quadratic U-shape ($\hat{\gamma}_5 = +0.0151$, $p = 0.017$) does not survive log-IO, kink-IO, or a flat twenty-bin diagnostic, so the discrete-tercile concentration is the headline functional-form-free characterization and the U-shape is parametric corroboration only.

8 Conclusion

A large literature documents that less financially literate investors trade systematically worse, and a separate literature documents that retail demand is geographically concentrated and moves prices. Together they predict that

state-level investor literacy should moderate the return anomalies that depend on retail mispricing. Aggregate tests deliver weak signals; the contribution of this paper is to show that the literacy mechanism is not absent but hidden by aggregation, and to characterize precisely where in the cross-section — and under what conditions on ownership and size — it operates.

The data identify an interaction zone: the literacy channel leaves a price footprint only where retail demand is large enough to move prices and institutional activity is sufficient to translate it into return differences detectable across states. Where neither holds (micro-caps held mostly by retail) mispricing accumulates without loading on state literacy; where institutions dominate (large high-IO firms) arbitrage neutralizes the local-state channel. In the full stable-HQ panel the zone falls in mid-IO; restricting to larger firms relocates it to low-IO, and the cross-state literacy gradient migrates with it, while firms outside the zone show no gradient. This migration — the moderation moving with sample composition — is the central new piece of evidence; it identifies the interaction-zone reading without an instrument, which is fortunate because the natural candidate (state K-12 personal-finance mandates) fails its first-stage relevance test for documented reasons. The result rests on OLS under conditional unconfoundedness, with the migration providing the predicted-signature confirmation an instrument would otherwise supply.

Three honest bounds discipline the result: the U-shape clears five percent on a quadratic form only (the discrete-tercile contrast is the functional-form-free claim); the point-in-time literacy reassignment collapses the headline

magnitude roughly twenty-six fold, leaving the stable-headquarters subsample as the population for which the characterization holds; and the within-mid-IO IV-decile decomposition rejects the textbook limits-to-arbitrage reading, so the supported microfoundation is identified through size and ownership conditioning, not the IV margin. Whether other state-level investor traits — participation rates, brokerage density, retail-attention measures — operate through a similar interaction zone is a natural next question, and the size-restriction migration test developed here provides a portable signature for testing it.

A Variable construction detail

Extended robustness material is in the Internet Appendix: the Herfindahl-Hirschman alternative (IA M), the Korniotis-Kumar business-cycle block (IA O), the difference-in-differences secondary design (IA T), the full PIT literacy reassignment (IA A), the relocater-vs-stable-HQ characteristics comparison (IA G), the Thomson Reuters s34 panel construction (IA P), and alternative literacy constructions (IA L).

NFCS Big Five share-correct score. The state financial-literacy measure is the share of NFCS respondents in a state-wave who answer the FINRA Big Five literacy questions correctly. Correct-answer keys from the FINRA codebook: M6=1, M7=3, M8=2, M9=2, M10=1. An earlier build coded all

five questions with a single key; the corrected keys are used throughout, and the Table 1 state-literacy rankings are post-correction. State-wave scores are forward-filled to monthly frequency between waves, then z-scored within calendar month to form `literacy_z`.

Idiosyncratic volatility. Standard deviation of daily Fama-French three-factor residuals over a rolling twelve-month window ending in the observation month; z-scored within calendar month before entering the interaction terms.

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